Behavioural, Functional, and Non-Functional Contracts for Dynamic Selection of Services

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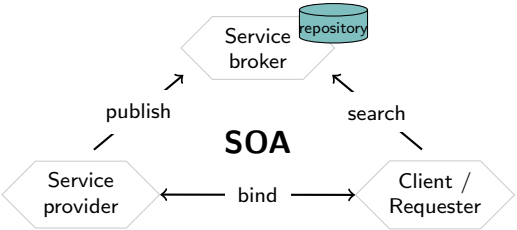
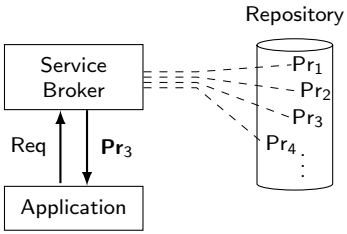
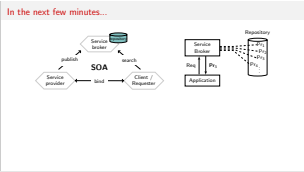
Work partly supported by the PRIN 2022 PNRR project DeLiCE (F53D23009130001)

In the next few minutes...







I'll present a formal framework motivated by dynamic discovery&binding of service

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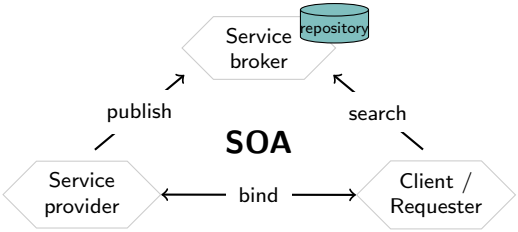
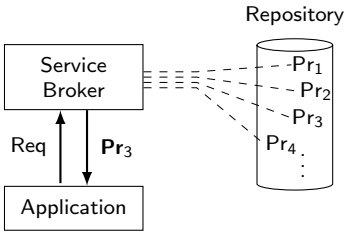
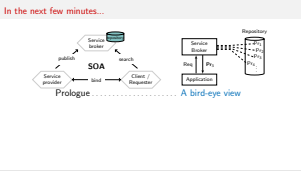




Prologue . . . . . A bird-eye view



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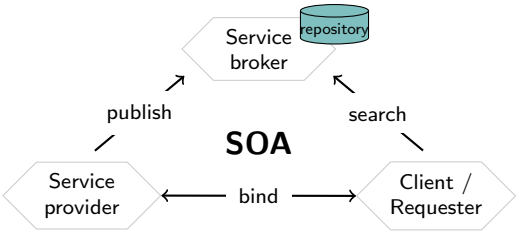

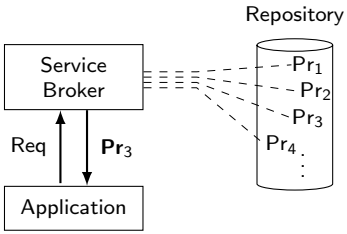

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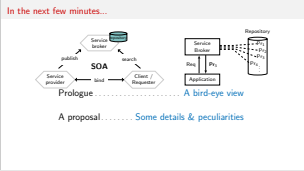

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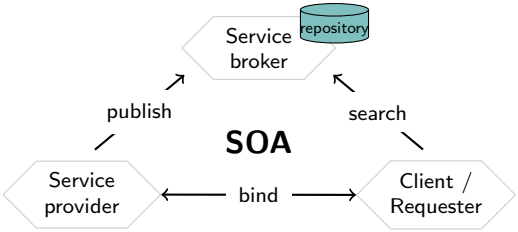

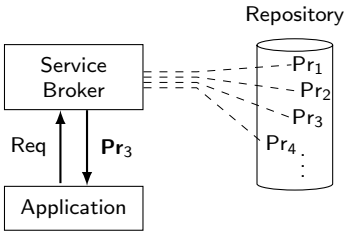

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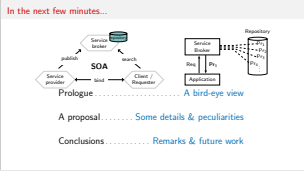

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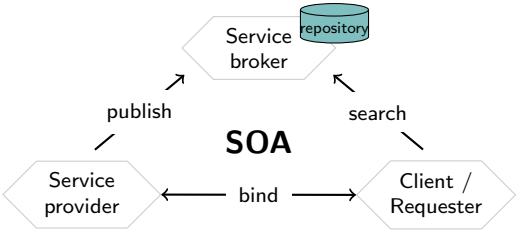

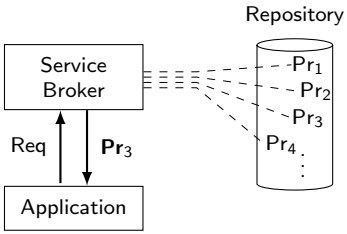

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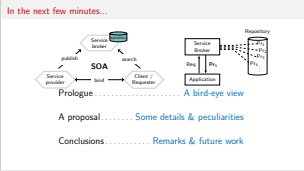

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[Prologue]

This talk in 1 slide

The problem

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Using contracts in SOC is not new
We strive for a uniform framework (to be tool supported in the future)

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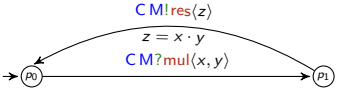

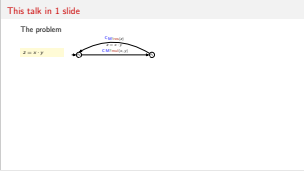

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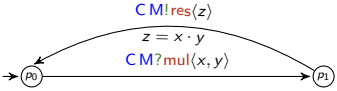

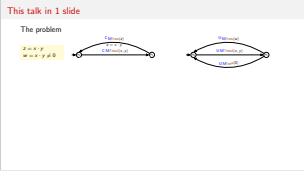

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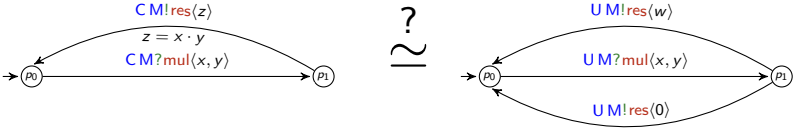

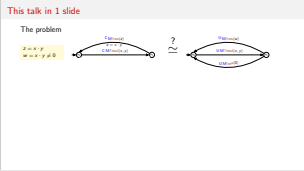

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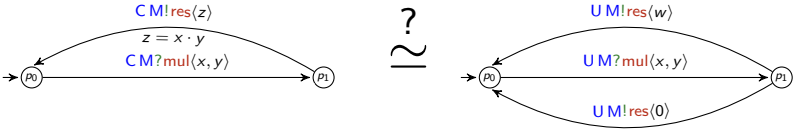
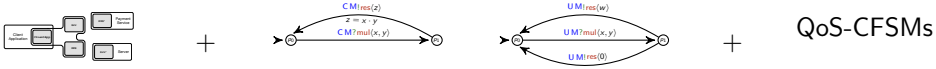
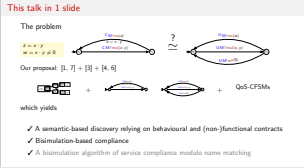


Our proposal: [1, 7] + [3] + [4, 6]



which yields

- ✓ A semantic-based discovery relying on behavioural and (non-)functional contracts
- ✓ Bisimulation-based compliance
- ✓ A bisimulation algorithm of service compliance modulo name matching



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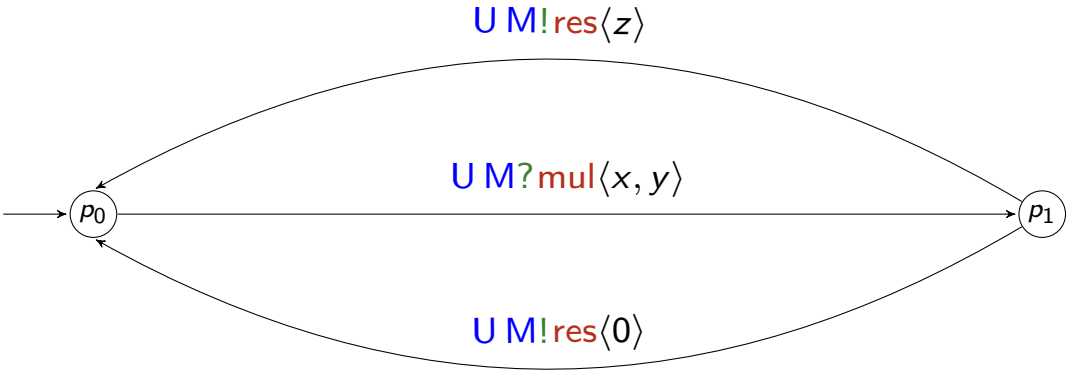
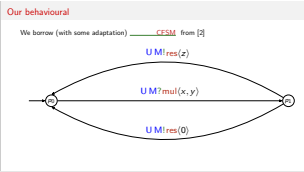
[Preliminaries]

Our behavioural

We borrow (with some adaptation) CFSM from [2]





Our behavioural and functional contracts

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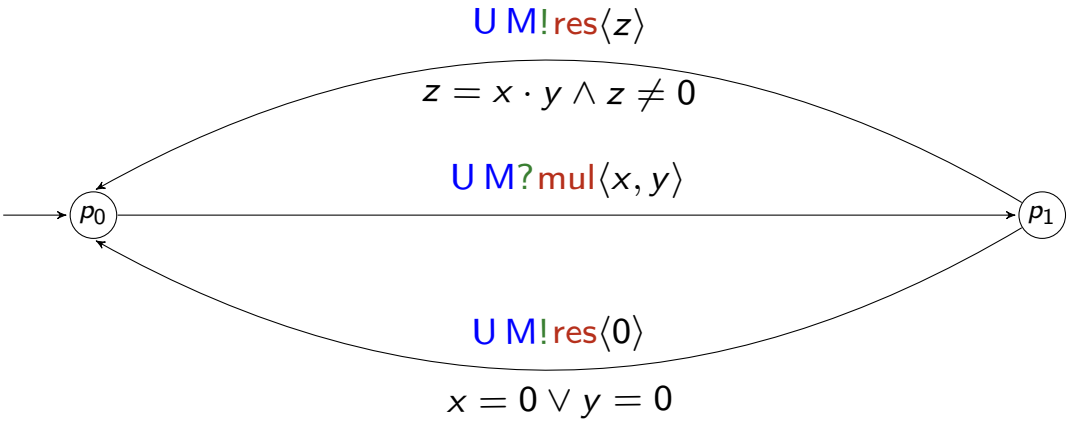

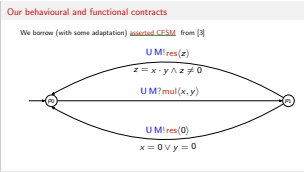

Our non-functional contracts

A variant of CFSMs yields our behavioural contracts; our non-functional contracts are

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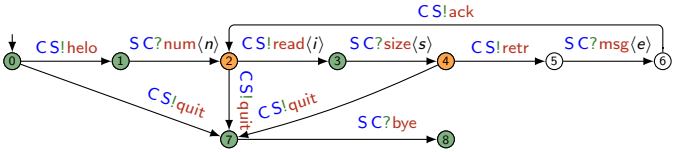

$$\Gamma_{\text{Low}} = \{t \leq 0.01,\ c \leq 0.01,\ m \leq 0.01\}$$

$$\Gamma_{\text{DB}} = \{t \leq 3 \implies (\exists x)(0.5 \leq x \leq 1 \wedge c = t \cdot x),\ t > 3 \implies c = 10,\ m \leq 5\}$$

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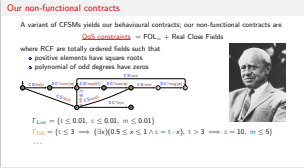

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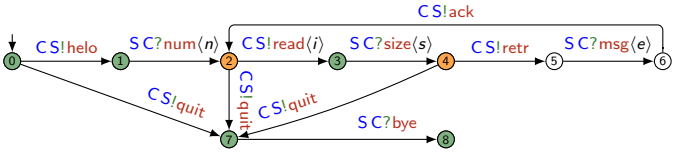

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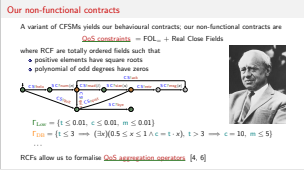
RCFs allow us to formalise QoS aggregation operators [4, 6]



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[A proposal]

Extended CFSMs

An extended CFSM (e-CFSM for short) is a tuple $\langle M, F, \mathsf{qos}, \mathsf{asrt} \rangle$ where:

- $M = \langle Q, q_0, \rightarrow \rangle$ is a CFSM with $F \subseteq Q$ the set of final states ,
- asrt maps transitions of M to $\mathrm{FOL}_{=}$
- $\mathsf{qos} : Q \rightarrow \mathcal{C}$ maps states of M to set of QoS constraints

An extended communicating system is a map $(M_{\mathsf{A}})_{\mathsf{A} \in \mathcal{P}}$ assigning an A-local e-CFSM M_{A} to each $\mathsf{A} \in \mathcal{P}$.

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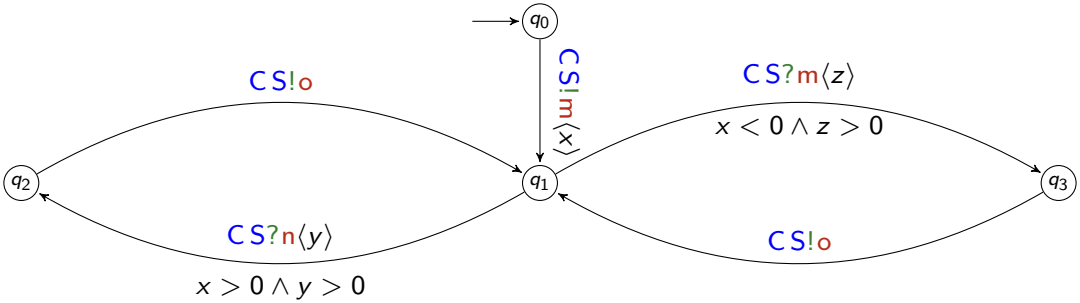

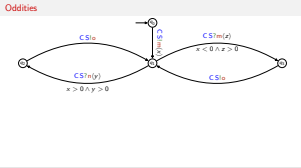

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do we know that $y > 0$ in t?

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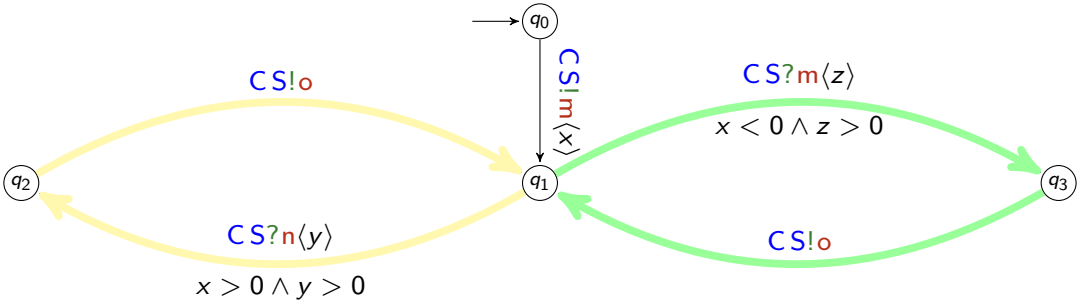

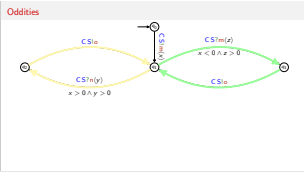

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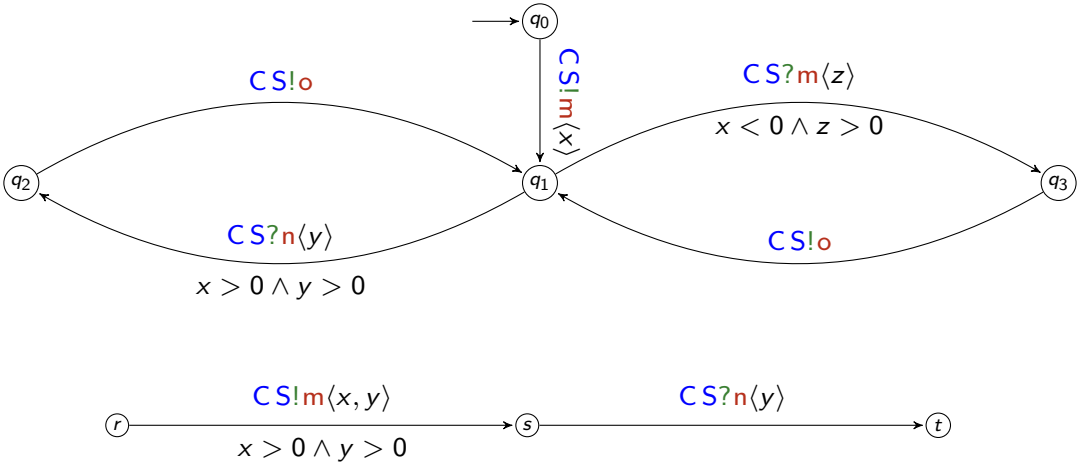

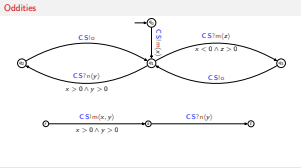

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Knowledge

The residual $\phi \mathbin{\overline{\wedge}} l$ of an assertion ϕ after l is $\bot$ unless

$$
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p(x_1, \ldots, x_n) \mathbin{\overline{\wedge}} l &= p(x_1, \ldots, x_n) && \text{if } var(l) \cap \{x_1, \ldots, x_n\} = \emptyset \\
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(\phi_1 \vee \phi_2) \mathbin{\overline{\wedge}} l &= (\phi_1 \mathbin{\overline{\wedge}} l) \vee (\phi_2 \mathbin{\overline{\wedge}} l) && \text{if } \phi_1 \mathbin{\overline{\wedge}} l \neq \bot \text{ and } \phi_2 \mathbin{\overline{\wedge}} l \neq \bot \\
((\exists x)\phi) \mathbin{\overline{\wedge}} l &= (\exists x)(\phi \mathbin{\overline{\wedge}} l) && \text{if } x \notin var(l) \text{ and } \phi \mathbin{\overline{\wedge}} l \neq \bot \\
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[Conclusions]

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- bisimulation checking algorithm module name matching

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